

ON GENERALIZED MULTIVARIATE DISTRIBUTIONS

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1. Introduction

In this discussion we introduce generalized multivariate distributions from which most of the known multivariate distributions are obtained as special cases. Some special and interesting properties enjoyed by these generalized multivariate distributions are pointed out. Many distributions in multivariate analysis can be expressed in a form involving hypergeometric functions (${}_rF_q$) of matrix argument. The hypergeometric functions ${}_rF_q$ is defined by Constantine (1963).

2. A generalized distribution which includes the central and non-central Wishart distribution

Consider the probability distribution with the probability density function (p.d.f.)

$$(2.1) \quad f(X) = \frac{1}{\Gamma_p(n) {}_rF_q(a_1, \dots, a_r; n; b_1, \dots, b_q; \theta)} \cdot \exp[-\text{tr } X] |X|^{n - \frac{1}{2}(p+1)} {}_rF_q(a_1, \dots, a_r; b_1, \dots, b_q; X\theta)$$

for $X : \text{pxp} > 0$ and $f(X) = 0$ elsewhere, where $\theta : \text{pxp}$ is a positive definite matrix. The parameters $a_i, b_j; i=1, 2, \dots, r, j=1, 2, \dots, q$ are restricted to take those values for which $f(X)$ is positive.

The following cases are obtained.

(i) *The central Wishart distribution*

In (2.1), putting $\theta=0$, we get

$$f(X) = \frac{1}{\Gamma_p(n)} \exp[-\text{tr } X] |X|^{n - \frac{1}{2}(p+1)} \quad \text{for } X > 0$$

where the covariance matrix $\Sigma=1$ in the original distribution.

(ii) *The non-central Wishart distribution (Constantine (1963) and Herz (1955))*

In (2.1), putting $r=0, q=1$ and $b_1=n$, we get

$$f(X) = \frac{1}{\Gamma_p(n)} \exp[-\text{tr } \theta] \exp[-\text{tr } X] |X|^{n - \frac{1}{2}(p+1)} {}_0F_1(n; X\theta)$$

where θ is the non-centrality parameter. We now have the following properties.

(a) The moment generating function (m.g.f.) of (2.1) is given by

$$M(T) = E[\exp\{\text{tr } TX\}]$$

By using the integral (Constantine (1963)),

$$(2.2) \quad \int_{X \geq 0} \exp[-\text{tr } TX] |X|^{n - \frac{1}{2}(p+1)} C_K(X\theta) dX = \Gamma(n, K) \frac{C_K(T^{-1}\theta)}{|T|^n}$$

we get the m.g.f. as

$$(2.3) \quad M(T) = \frac{r+1 F_q(a_1, \dots, a_r, n; b_1, \dots, b_q; (I-T)^{-1}\theta)}{r+1 F_q(a_1, \dots, a_r, n; b_1, \dots, b_q; \theta)} |1-T|^{-n}$$

(b) The h -th moment of $|X|$ can be found from (2.1) by using the fact that $\int f(X) dX = 1$ as

$$E|X|^h = \frac{\Gamma_p(n+h)}{\Gamma_p(n)} \frac{r+1 F_q(a_1, \dots, a_r, n+h; b_1, \dots, b_q; \theta)}{r+1 F_q(a_1, \dots, a_r, n; b_1, \dots, b_q; \theta)}$$

(c) The distribution of $\text{tr } X$ will be defined by inverting the Laplace transform

$$g(s) = E[\exp\{-s(\text{tr } X)\}]$$

of its density function.

From (2.3)

$$g(s) = \frac{r+1 F_q(a_1, \dots, a_r, n; b_1, \dots, b_q; (1+s)^{-1}\theta)}{r+1 F_q(a_1, \dots, a_r, n; b_1, \dots, b_q; \theta)} (1+s)^{-np}$$

Using the Laplace inverse i.e.

$$f(\text{tr } X) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \exp[s(\text{tr } X)] g(s) ds$$

and integrating term by term with respect to s , we have the following integral (Erdélyi et al. (1954)):

$$\begin{aligned} & \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \exp[s(\text{tr } X)] (1+s)^{-(k+np)} ds \\ &= \frac{1}{\Gamma(k+np)} \exp[-\text{tr } X] (\text{tr } X)^{k+np-1} \end{aligned}$$

Then the density function of $\text{tr } X$ is given by

$$f(\text{tr } X) = \frac{\exp[-\text{tr } X J]}{r+1 F_q(a_1, \dots, a_r, n; b_1, \dots, b_q; \theta)} \cdot \sum_{k=0}^{\infty} \sum_K \frac{(a_1)_K \dots (a_r)_K (n)_K}{(b_1)_K \dots (b_q)_K} \frac{(\text{tr } X)^{k+np-1}}{\Gamma(k+np) k!} C_K(\theta)$$

(d) To calculate the moments of $\text{tr } X$ we note that (Constantine (1966))

$$(\text{tr } X)^j = \sum_J C_J(X)$$

Therefore,

$$E(\text{tr } X)^j = \sum_J \int_{X \geq 0} C_J(X) f(X) dX$$

The integration with respect to X can be carried out using (De Waal (1968))

$$\int_{X \geq 0} \exp[-\text{tr } X] |X|^{n-\frac{1}{2}(p+1)} C_K(X \theta) C_J(X) dX$$

$$= \Gamma_p(n) \frac{C_K(\theta)}{C_K(I_p)} \sum_{\delta} g_{K,J}^{\delta}(n)_{\delta} C_{\delta}(I_p)$$

where $C_K(X) C_J(X) = \sum_{\delta} g_{K,J}^{\delta} C_{\delta}(X)$. The result is

$$E(\text{tr } X)^j = \frac{1}{r+1 F_q(a_1, \dots, a_r, n; b_1, \dots, b_q; \theta)} \cdot \sum_{k=0}^{\infty} \sum_K \sum_J \sum_{\delta} \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} g_{K,J}^{\delta}(n)_{\delta} \frac{C_K(\theta) C_{\delta}(I_p)}{C_K(I_p)}$$

3. A generalized distribution which includes the central and non-central multivariate beta distribution

Consider the probability distributions with the p.d.f.'s

$$(3.1) \quad f(X) = \frac{\Gamma_p(m+n)}{\Gamma_p(m) \Gamma_p(n) r+1 F_q(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n; \theta)} \cdot |X|^{m-\frac{1}{2}(p+1)} |I-X|^{n-\frac{1}{2}(p+1)} {}_rF_q(a_1, \dots, a_r; b_1, \dots, b_q; \theta X)$$

for $0 < X < I$ and $f(X) = 0$ elsewhere, and

$$(3.2) \quad f(Y) = \frac{\Gamma_p(m+n)}{\Gamma_p(m)\Gamma_p(n)} \frac{\Gamma_p(m+h)}{r+1 F_{q+1}(a_1, \dots, a_r, n; b_1, \dots, b_q, m+n; \theta)} \cdot |I+Y|^{-(m+n)} |Y|^{m-\frac{1}{2}(p+1)} {}_rF_q(a_1, \dots, a_r; b_1, \dots, b_q; \theta(I+Y)^{-1})$$

for $Y : p \times p > 0$ and $f(Y) = 0$ elsewhere. The parameters are restricted to take those values for which $f(X)$ and $f(Y)$ are positive. The distributions with p.d.f.'s (3.1) and (3.2) include under certain conditions the central (Anderson (1958) and Olkin and Rubin (1964)) and non-central (De Waal (1968) and De Waal (1969)) multivariate beta distributions.

(a) The h -th moments can be found from (3.1) and (3.2) by using the fact that $\int f(X) dX = 1$ and $\int f(Y) dY = 1$. Hence

$$E|X|^h = \frac{\Gamma_p(m+n)\Gamma_p(m+h)}{\Gamma_p(m)\Gamma_p(m+n+h)} \cdot \frac{{}_r+1 F_{q+1}(a_1, \dots, a_r, m+h; b_1, \dots, b_q, m+n+h; \theta)}{{}_r+1 F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n; \theta)},$$

$$E|I-X|^h = \frac{\Gamma_p(m+n)\Gamma_p(n+h)}{\Gamma_p(n)\Gamma_p(m+n+h)} \cdot \frac{{}_r+1 F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n+h; \theta)}{{}_r+1 F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n; \theta)},$$

$$E|Y|^h = \frac{\Gamma_p(n-h)\Gamma_p(m+h)}{\Gamma_p(n)\Gamma_p(m)} \cdot \frac{{}_r+1 F_{q+1}(a_1, \dots, a_r, n-h; b_1, \dots, b_q, m+n; \theta)}{{}_r+1 F_{q+1}(a_1, \dots, a_r, n; b_1, \dots, b_q, m+n; \theta)},$$

$$E|I+Y|^{-h} = \frac{\Gamma_p(m+n)\Gamma_p(n+h)}{\Gamma_p(n)\Gamma_p(m+n+h)} \cdot \frac{{}_r+1 F_{q+1}(a_1, \dots, a_r, n+h; b_1, \dots, b_q, m+n+h; \theta)}{{}_r+1 F_{q+1}(a_1, \dots, a_r, n; b_1, \dots, b_q, n+h; \theta)}$$

$$\text{and } E | Y(I+Y)^{-1} h = \frac{\Gamma_p(m+n) \Gamma_p(m+h)}{\Gamma_p(m) \Gamma_p(m+n+h)} \cdot$$

$$\frac{{}_r+1F_{q+1}(a_1, \dots, a_r, n; b_1, \dots, b_q, m+n+h; \theta)}{{}_r+1F_{q+1}(a_1, \dots, a_r, n; b_1, \dots, b_q, m+n; \theta)}$$

(b) The m.g.f. of $\text{tr } X = \text{tr } \Delta$ is given by

$$(3.3) \quad M(a) = E[\exp(\text{tr } \Delta a)]$$

$$= \sum_j \sum_J a^j E[C_J(\Delta)] \frac{1}{j!}$$

where $\Delta = \text{diag.}(x_1, \dots, x_p)$.

$$E[C_J(\Delta)] = \pi^{\frac{1}{2}p^2} \cdot \frac{1}{\Gamma_p(\frac{1}{2}p)} \frac{\Gamma_p(n+m)}{\Gamma_p(n) \Gamma_p(m) {}_r+1F_{q+1}(a_1, \dots, a_r, n; b_1, \dots, b_q, n+m; \theta)} \cdot$$

$$(3.4) \quad \sum_{k=0}^{\infty} \sum_K \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} \frac{C_K(\theta)}{C_K(I_p) k!} \cdot$$

$$\int_0^I |\Delta|^{n-\frac{1}{2}(p+1)} |I-\Delta|^{m-\frac{1}{2}(p+1)} a_p(\Delta) C_K(\Delta) C_J(\Delta) d\Delta \quad (\text{Constantine (1963)})$$

$$\text{where } a_p(\Delta) = \prod_{i < j}^p (x_i - x_j)$$

Using the integral Sugiyama (1966)

$$\begin{aligned} & \int_0^I |\Delta|^{n-\frac{1}{2}(p+1)} |I-\Delta|^{m-\frac{1}{2}(p+1)} a_p(\Delta) C_K(\Delta) C_J(\Delta) d\Delta \\ &= \sum_{\delta} g_{K,J}^{\delta} \frac{\Gamma_p(\frac{1}{2}p) \Gamma_p(m) \Gamma_p(n, \delta) C_{\delta}(I_p)}{\pi^{\frac{1}{2}p^2} \Gamma_p(n+m, \delta)} \end{aligned}$$

it follows from (3.3) and (3.4) that

$$\begin{aligned} M(a) &= \sum_{j=0}^{\infty} \sum_J \sum_{k=0}^{\infty} \sum_K \sum_{\delta} \frac{a^j}{j!} g_{K,J}^{\delta} \frac{(a_1)_K \dots (a_r)_K (n)_{\delta} C_K(\theta) C_{\delta}(I_p)}{(b_1)_K \dots (b_q)_K (n+m)_{\delta} C_K(I_p) k!} \\ &\quad \cdot \frac{1}{{}_r+1F_{q+1}(a_1, \dots, a_r, n; b_1, \dots, b_q, n+m; \theta)} \end{aligned}$$

(c) Consider

$$(3.5) \quad g(s) = E [\exp[-s(\text{tr } Y)]]$$

$$= \int_{Y>0} \exp[-s(\text{tr } Y)] f(Y) dY$$

Making the transformation $\theta \rightarrow H'\theta H$ in (3.2) and integrating over $\text{HeO}(p)$ using

$$\begin{aligned} & \int_{O(p)} r F_q(a_1, \dots, a_r; b_1, \dots, b_q; H'\theta H(I+Y)^{-1}) d(H) \\ &= r F_q(a_1, \dots, a_r; b_1, \dots, b_q; \theta, (I+Y)^{-1}), \text{ and (Troskie (1968))} \\ & |I+Y|^{-(n+m)} (n+m)_K C_K((I+Y)^{-1}) \\ &= \sum_{j=0}^{\infty} \sum_j (-1)^j \sum_{\delta} (n+m)_{\delta} g_{K,J}^{\delta} \frac{C_J(Y) C_{\delta}(I_p)}{C_J(I_p) j!} \end{aligned}$$

we obtain from (3.5) and (2.2)

$$\begin{aligned} g(s) &= \frac{\Gamma_p(n+m)}{\Gamma_p(n) \Gamma_p(m) r+1 F_{q+1}(a_1, \dots, a_r, n; b_1, \dots, b_q, n+m; \theta)} \\ & \sum_{j=0}^{\infty} \sum_j \sum_{k=0}^{\infty} \sum_K (-1)^j \frac{(n+m)_{\delta}}{(n+m)_K} g_{K,J}^{\delta} \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} \frac{C_K(Z) C_{\delta}(I_p)}{C_K(I_p) k! j!} \\ & \cdot \Gamma_p(m, J) s^{-mp-j} \end{aligned}$$

Using the Laplace inverse, i.e.

$$f(\text{tr } Y) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \exp[-s(\text{tr } Y)] g(s) ds$$

and integrating term by term, we have the following integral (Erdélyi et. al. (1954), p. 238 (1))

$$\begin{aligned} & \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \exp[-s(\text{tr } Y)] s^{-(mp+j)} ds \\ &= (\text{tr } Y)^{mp+j-1} \frac{1}{\Gamma(mp+j)} \end{aligned}$$

Then the density function of $\text{tr } Y$ is given by

$$f(\text{tr } Y) = \frac{\Gamma_p(n+m)}{\Gamma_p(n) \Gamma_p(m) r+1 F_{q+1}(a_1, \dots, a_r, n; b_1, \dots, b_q, n+m; \theta)}$$

$$\cdot \sum_{j=0}^{\infty} \sum_{j=1}^{\infty} \sum_{k=0}^{\infty} \sum_{k=1}^{\infty} (-1)^j \frac{(n+m)_\delta}{(n+m)_K} g_{K,J}^\delta \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} \cdot \frac{C_K(Z) C_\delta(I_p)}{C_K(I_p) k! j!} \frac{\Gamma_p(m, J)}{\Gamma(m p + j)} (sp Y)^{mp + j - 1}$$

4. A generalized distribution which includes the central and non-central multivariate Dirichlet distribution

As extension of (3.1) and (3.2) consider the probability distributions with p.d.f.'s

$$(4.1) \quad f(X_{(1)}, X_{(2)}, \dots, X_{(k)}) \\ = \frac{\Gamma_p(m+n)}{\prod_{j=1}^k \Gamma_p(m_j) \Gamma_p(n) {}_{r+1}F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n; \theta)} \cdot \prod_{j=1}^k |X_{(j)}|^{m_j - \frac{1}{2}(p+1)} \left| 1 - \sum_{j=1}^k X_{(j)} \right|^{n - \frac{1}{2}(p+1)} \cdot {}_{r+1}F_q(a_1, \dots, a_r; b_1, \dots, b_q; \theta \sum_{j=1}^k X_{(j)})$$

for $I - \sum_{j=1}^k X_{(j)} > 0$, $0 < X_{(j)} < 1$ and $f(X_{(1)}, \dots, X_{(k)}) = 0$ elsewhere where $m = \sum_{j=1}^k m_j$ and

$$(4.2) \quad f(Y_{(1)}, Y_{(2)}, \dots, Y_{(k)}) \\ = \frac{\Gamma_p(m+n)}{\prod_{j=1}^k \Gamma_p(m_j) \Gamma_p(n) {}_{r+1}F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n; \theta)} \cdot \prod_{j=1}^k |Y_{(j)}|^{m_j - \frac{1}{2}(p+1)} \left| 1 + \sum_{j=1}^k Y_{(j)} \right|^{-(m+n)} \cdot {}_{r+1}F_q(a_1, \dots, a_r; b_1, \dots, b_q; \theta (1 + \sum_{j=1}^k Y_{(j)})^{-1})$$

for $Y_{(j)} > 0$ and $f(Y_{(1)}, \dots, Y_{(k)}) = 0$ elsewhere.

The parameters are assumed to be such that the p.d.f.'s are positive. The distributions with p.d.f.'s (4.1) and (4.2) include under certain conditions the central (Olkin and Rubin (1964) and Troskie (1966)) and non-central (De Waal (1968)) multivariate Dirichlet distributions.

From the definitions (4.1) and (4.2) we have the following moments:

$$E\left(\prod_{j=1}^k |X_{(j)}|^{h_j}\right) = \frac{\Gamma_p(m+n) \prod_{j=1}^k \Gamma(m_j+h_j)}{\prod_{j=1}^k \Gamma_p(m_j) \Gamma_p(m+n+h)} \cdot \frac{r+1 F_{q+1}(a_1, \dots, a_r, m+h; b_1, \dots, b_q, m+n+h; \theta)}{r+1 F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n; \theta)}$$

where $h = \sum_{j=1}^k h_j$,

$$E\left|1 - \sum_{j=1}^k X_{(j)}\right|^h = \frac{\Gamma_p(m+n) \Gamma_p(n+h)}{\Gamma_p(n) \Gamma_p(m+n+h)} \cdot \frac{r+1 F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n+h; \theta)}{r+1 F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n; \theta)},$$

$$E\left(\prod_{j=1}^k |Y_{(j)}|^{h_j}\right) = \frac{\prod_{j=1}^k \Gamma_p(m_j+h_j) \Gamma_p(n-h)}{\prod_{j=1}^k \Gamma_p(m_j) \Gamma_p(n)} \cdot \frac{r+1 F_{q+1}(a_1, \dots, a_r, m+h; b_1, \dots, b_q, m+n; \theta)}{r+1 F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n; \theta)}$$

and

$$E\left|1 + \sum_{j=1}^k Y_{(j)}\right|^h = \frac{\Gamma_p(m+n) \Gamma_p(n+h)}{\Gamma_p(n) \Gamma_p(m+n+h)} \cdot \frac{r+1 F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n+h; \theta)}{r+1 F_{q+1}(a_1, \dots, a_r, m; b_1, \dots, b_q, m+n; \theta)}$$

5. Distributions of random matrices with random matrix parameter

We consider the cases where the conditional distribution, assumed to be of a generalized multivariate gamma type, contains a matrix parameter with an a priori generalized multivariate distribution.

Theorem 5.1

Let the conditional distribution of a random matrix X , given the matrix parameter P , be a multivariate gamma with p.d.f.

$$(5.1) \quad f(X|P) = \frac{1}{\Gamma_p(m)} |P|^m |X|^{m-1/2(p+1)} \exp[-\text{tr } PX]$$

for $X > 0$ and $f(X|P) = 0$ elsewhere. Let the marginal distribution of $P: p \times p$ be given by (2.1) with p.d.f.

$$g(P) = \frac{1}{\Gamma_p(n) \Gamma_p(r+1) F_q(a_1, \dots, a_r; n; b_1, \dots, b_q; \theta)} \cdot \exp[-\text{tr } P] |P|^{n-1/2(p+1)} F_q(a_1, \dots, a_r; b_1, \dots, b_q; P\theta)$$

Then the unconditional distribution of X is given by the p.d.f.

$$(5.2) \quad h(X) = \frac{\Gamma_p(m+n)}{\Gamma_p(m) \Gamma_p(n) \Gamma_p(r+1) F_q(a_1, \dots, a_r; n; b_1, \dots, b_q; \theta)} \cdot |X|^{m-1/2(p+1)} |I+X|^{-(m+n)} \cdot \Gamma_p(r+1) F_q(a_1, \dots, a_r; m+n; b_1, \dots, b_q; \theta) (I+X)^{-1}$$

for $X > 0$, where n, m, a_i, b_i and θ are additional parameters.

Proof

The unconditional distribution is given by,

$$h(X) = \int_{P>0} f(X|P) g(P) dP$$

From (5.1) and (5.2) and using (2.2) the theorem is proved.

Theorem 5.2

Let the random matrices $X_{(1)}, \dots, X_{(k)}$ have a conditional distribution with p.d.f.

$$f(X_{(1)}, \dots, X_{(k)}|P) = \frac{1}{\prod_{j=1}^k \Gamma_p(m_j)} |P|^{m} |X_{(j)}|^{m_j-1/2(p+1)} \exp[-\text{tr}(P \sum_{j=1}^k X_{(j)})]$$

for $X_{(i)} > 0$ and $f(X_{(1)}, \dots, X_{(k)}|P) = 0$ elsewhere, where $m = \sum_{j=1}^k m_j$

Let the marginal distribution of $P: p \times p$ be given by (5.2). Then the unconditional distribution of $X_{(1)}, \dots, X_{(k)}$ is given by the p.d.f.

$$h(X_{(1)}, \dots, X_{(k)}) = \frac{\Gamma_p(m+n)}{\prod_{j=1}^k \Gamma_p(m_j) \Gamma_p(n) \Gamma_p(r+1) F_q(a_1, \dots, a_r; n; b_1, \dots, b_q; \theta)}$$

$$\cdot \pi \prod_{j=1}^k |X_{(j)}|^{m_j - \frac{1}{2}(p+1)} |I + \sum_{j=1}^k X_{(j)}|^{-(n+m)} \cdot$$

$$\cdot r + I F_q(a_1, \dots, a_r, m+n; b_1, \dots, b_q; \theta \left(I + \sum_{j=1}^k X_{(j)} \right)^{-1})$$

where n, m_j, a_i, b_i and θ are additional parameters.

The proof follows as in theorem 5.1.

The author wishes to thank Prof. D.J. de Waal for his helpful suggestions and discussions.

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Manuscript received 25th February 1971; revised 25th May 1971.